

## ARTICLE TYPE

# Robust Regression with Student's $T$ : The Role of Degrees of Freedom

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## Abstract

Linear regression estimators are known to be sensitive to outliers, and one alternative to obtain a robust and efficient estimator of the regression parameter is to model the error with Student's  $t$  distribution. In this article, we compare estimators of the degrees of freedom parameter in the  $t$  distribution using frequentist and Bayesian methods, and then study properties of the corresponding estimated regression coefficient. We also include the comparison with some recommended approaches in the literature, including fixing the degrees of freedom and robust regression using the Huber loss. Our extensive simulations on both synthetic and real data demonstrate that estimating the degrees of freedom via the adjusted profile log-likelihood approach yields regression coefficient estimators with high accuracy, performing comparably to the maximum likelihood estimator where the degrees of freedom are fixed at their true values. These findings provide a detailed synthesis of  $t$ -based robust regression and underscore a key insight: the proper calibration of the degree of freedom is as crucial as the choice of the robust distribution itself for achieving optimal performance.

## KEY WORDS

outliers, Jeffreys' prior, adjusted profile likelihood

## 1 | INTRODUCTION

Robustness is a fundamental concern in statistical inference. The presence of outliers can distort parameter estimates and degrade the reliability of inference. Some methods that have been developed for robust inference include  $M$ -estimation, with estimating functions designed to downweigh outliers (Huber, 1973),  $L$ -estimation, based on linear combinations of order statistics (Koenker & Portnoy, 1987), and  $R$ -estimation, based on ranks (Huber & Ronchetti, 2009, Chapter 3).

In this paper we consider robust estimation in the linear regression model

$$y = X\beta + \sigma\epsilon, \quad (1)$$

where  $y = (y_1, \dots, y_n)$  is a vector of responses,  $X$  is an  $n \times p$  matrix of covariates with  $i^{\text{th}}$  row  $x_i^T$ , and  $\epsilon = (\epsilon_1, \dots, \epsilon_n)$  are unobserved random terms of mean zero and variance 1, independent of  $y$  and  $X$ . Following several authors (Lange et al., 1989; Zellner, 1976; Singh, 1988; Fernández & Steel, 1999), we assume  $\epsilon_i, i = 1, \dots, n$  are independent and identically distributed as Student's  $t$  with  $\nu$  degrees of freedom.

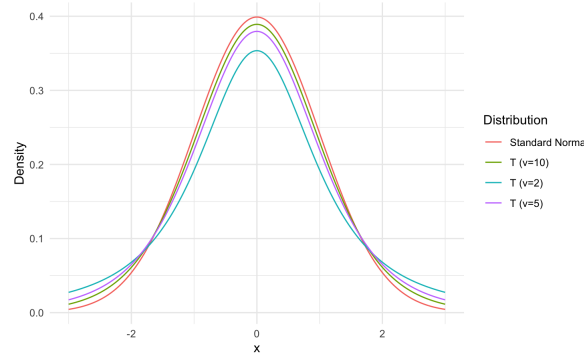
Under this assumption, the likelihood function for (1) is

$$L(\beta, \sigma, \nu; y) = \frac{\Gamma^n((\nu + 1)/2)}{\sigma^n(\pi\nu)^{n/2}\Gamma^n(\nu/2)} \prod_{i=1}^n \left\{ 1 + \frac{(y_i - x_i^T\beta)^2}{\nu\sigma^2} \right\}^{-(\nu+1)/2}. \quad (2)$$

Whether the covariates  $X$  are fixed or random, inference is based on the conditional density of  $y$  given  $X$ , as is usual in regression problems. The estimates derived from (2) may be expected to be more robust than least-squares estimates, as the  $t$ -distribution

has heavier tails than the normal distribution, especially for smaller values of  $\nu$ ; see Figure 1. As  $\nu \rightarrow \infty$ , the  $t$ -density function converges to the normal density function.

The ordinary least squares estimator of  $\beta$  has maximum efficiency under the normal model, but is also sensitive to outliers. Models with smaller values of  $\nu$  can accommodate heavier tails, but may be inefficient if the data suggests a larger value of  $\nu$ . Lange et al. (1989) suggested estimating  $\nu$  when  $n$  is large but choosing some fixed value if  $n$  is small. Liu & Rubin (1995) described an extended expectation-maximization (EM) algorithm for computing maximum likelihood estimates of the  $(\beta, \sigma, \nu)$ . Singh (1988) proposed a moment-based estimator of  $\nu$  for  $\nu > 4$ . Several authors have considered Bayesian inference for  $\nu$ , using various choices of noninformative priors, such as Jeffreys' prior (Fonseca et al., 2008; Fernández & Steel, 1999), a reference prior (He et al., 2021) and a uniform prior (Relles & Rogers, 1977).



**FIGURE 1** Student's  $t$ -distribution has heavier tails than the normal distribution, with the difference decreasing as  $\nu$  increases. Asymptotically, as  $\nu \rightarrow \infty$  the  $t$ -distribution converges to the normal distribution.

In Section 2 we compare frequentist and Bayesian approaches to the estimation of  $\nu$ . In Section 3 we focus on inference for  $\beta$ , using a two-stage procedure in which we first estimate the degrees of freedom parameter  $\nu$ , and then fix  $\nu$  at this estimated value in the likelihood function and maximize over  $\beta$  and  $\sigma$ . With large  $p$  we find that estimating  $\nu$  by maximizing the adjusted profile likelihood function gives mean-squared error for  $\beta$  comparable to that from minimizing Huber loss Huber (1973). In Section 4 we consider the robustness of our proposed methods in the presence of outliers.

## 2 | ESTIMATING THE DEGREES OF FREEDOM

### 2.1 | Likelihood Approaches

The first estimator of  $\nu$  we consider is the maximum likelihood estimator,  $\hat{\nu}$ , which maximizes the profile log-likelihood function

$$\ell_p(\nu; y) = \ell(\hat{\beta}_\nu, \hat{\sigma}_\nu, \nu; y), \quad (3)$$

where  $\ell(\theta) = \log L(\theta; y)$ , and  $(\hat{\beta}_\nu, \hat{\sigma}_\nu) = \arg \max_{(\beta, \sigma)} \ell(\beta, \sigma, \nu; y)$  are the constrained maximum likelihood estimates of  $(\beta, \sigma)$ . Maronna (1976) and Zellner (1976) showed that  $\hat{\beta}_\nu$  and  $\hat{\sigma}_\nu$  exist. The asymptotic properties of  $\hat{\nu}$  follow from classical likelihood theory:

$$\sqrt{n} j_p^{1/2}(\hat{\nu})(\hat{\nu} - \nu) \xrightarrow{d} N(0, 1), \quad n \rightarrow \infty,$$

where  $j_p(\nu) = -\ell_p''(\nu)$  can be computed from the full information matrix; see the supplementary material.

An adjusted profile log-likelihood function was suggested in Cox & Reid (1987) to adjust for errors in estimation of the nuisance parameters  $\lambda = (\beta, \sigma)$ ,

$$\ell_{\text{adj}}(\nu; y) = \ell_p(\nu; y) - \frac{1}{2} \log |j_{\lambda\lambda}(\nu, \hat{\lambda}_\nu)|, \quad (4)$$

where  $j_{\lambda\lambda}(\nu, \hat{\lambda}_\nu)$  is the sub-matrix of the observed Fisher information matrix that corresponds to the nuisance parameters. An expression for  $j_{\lambda\lambda}(\nu, \lambda)$  is given in the supplementary material. Our second likelihood-based estimate is  $\hat{\nu}_{adj}$ , the value that maximizes the adjusted profile log-likelihood function  $\ell_{adj}(\nu; y)$ .

## 2.2 | Bayesian Approaches

Jeffreys (1946) proposed non-informative priors based on the expected Fisher information matrix, which is

$$I(\beta, \sigma, \nu) = \begin{pmatrix} \frac{1}{\sigma^2} \cdot \frac{\nu+1}{\nu+3} \sum_{i=1}^n x_i x_i^T & 0 & 0 \\ 0 & \frac{2n}{\sigma^2} \cdot \frac{\nu}{\nu+3} & -\frac{2n}{\sigma} \cdot \frac{1}{(\nu+1)(\nu+3)} \\ 0 & -\frac{2n}{\sigma} \cdot \frac{1}{(\nu+1)(\nu+3)} & \frac{n}{4} \left[ \psi' \left( \frac{\nu}{2} \right) - \psi' \left( \frac{\nu+1}{2} \right) - \frac{2(\nu+5)}{\nu(\nu+1)(\nu+3)} \right] \end{pmatrix}. \quad (5)$$

Following Fonseca et al. (2008), we consider Jeffreys' so-called independence prior, in which the prior for the location parameter  $\beta$  is treated independently of that for  $(\sigma, \nu)$ . Jeffreys' rule priors are discussed in detail in Kass & Wasserman (1996). This leads to the prior

$$\pi_{IJ}(\beta) \propto 1, \quad \pi_{IJ}(\sigma, \nu) \propto \sigma^{-1} \left\{ \frac{\nu}{\nu+3} \right\}^{\frac{1}{2}} \left\{ \psi' \left( \frac{\nu}{2} \right) - \psi' \left( \frac{\nu+1}{2} \right) - \frac{2(\nu+3)}{\nu(\nu+1)^2} \right\}^{\frac{1}{2}}, \quad (6)$$

where the prior for  $(\sigma, \nu)$  is the determinant of the  $2 \times 2$  submatrix of (5). Using these priors we maximized the posterior distribution over  $(\beta, \sigma, \nu)$  and extracted the Bayesian MAP estimate  $\hat{\nu}_{Bayes}$ .

An alternative partially Bayes approach is to treat the profile likelihood function for  $\nu$  as a true likelihood function and use the prior  $\pi(\nu) \propto I_{\nu\nu}^{1/2}(\nu)$  to form a pseudo-posterior distribution. The resulting posterior distribution is proper, as proved in the supplementary material. The MAP estimate of  $\nu$  obtained from maximizing this pseudo-posterior distribution is denoted by  $\hat{\nu}_{Pseudo}$ .

Estimates of  $\nu$  from these objective functions are invariant to location-scale transformations. Specifically, for the linear model  $y_i = x_i\beta + \epsilon_i$ , transforming the response as  $y'_i = ay_i + x_i\vec{b}$  with  $a > 0$  and a constant vector  $\vec{b}$  shifts the log likelihoods and log pseudo-posterior by a constant, leaving the  $\nu$  optimization results unchanged. See the supplementary material for details.

Following Lange et al. (1989), we estimated the degrees of freedom  $\nu$  using the reparametrization  $\omega = 1/\nu$ . For the normal distribution, this parameterization corresponds to  $\omega \rightarrow 0$  or  $\nu \rightarrow \infty$ . When specifying priors, we applied the change of variable: if  $p(\nu)$  is the prior for  $\nu$ , then the prior for  $\omega$  is  $\pi(\omega) = p(1/\omega)/\omega^2$ .

## 2.3 | Simulations

In this section, we compare the performance of the four  $\nu$ -estimation approaches proposed above. We generated data from (1) with  $t$  errors for  $\epsilon_i$ , with degrees of freedom  $\nu = \{2, 5, 10\}$ ,  $\sigma = 1$ , and  $\beta \in \mathbb{R}^p$  with dimension  $p \in \{1, 2, 5, 10, 20, 40, 60, 80\}$ . The covariates were generated from a multivariate standard normal distribution  $\mathbf{N}_p(\mathbf{0}, \mathbf{I}_p)$ . The choice of the true values of  $\beta$  and  $\sigma$  is arbitrary as the inference for  $\nu$  is invariant to location-scale changes. The sample sizes were 300, 2500 and 4500 when  $\nu = 2, 5$ , and 10 (regardless of dimension  $p$ ) respectively.

The BFGS algorithm provided in  $\mathbb{R}$  (R Core Team, 2023) was used for all optimization steps in our simulations. This is a quasi-Newton method which iteratively approximates the Hessian of the objective function. The starting values of  $(\beta, \sigma)$  for constrained maximum likelihood estimation were the ordinary least squares estimate of  $\beta$  and usual residual mean square estimate of  $\sigma$ .

All computations were carried out in the  $\omega$  parameterization, with initial guess set to the true value of  $1/\nu$ , but results are reported for estimation of  $\nu$ . The covariates were fixed in each set of simulation, to reflect modelling the conditional expected value of  $y$  given  $x$ . We ran 500 simulations for each combination of  $\nu$  and  $p$ . The supplementary material provides more details about the choice of optimization details, including starting values, sample sizes and handling out-of-bound values.

Table 1 presents the root mean-square-error (RMSE) averaged over the simulations, for the four estimators of  $\nu$ , when the true value of  $\nu = 2$  and the sample size  $n = 300$ . For small values of  $p$  the maximum profile likelihood estimator  $\hat{\nu}$  has the smallest RMSE, but all four methods are comparable. For  $p = 60$  or 80, the adjusted profile likelihood estimator  $\hat{\nu}_{adj}$  has the smallest RMSE, and seems to adapt well to high-dimensional settings. The Bayesian estimators have similar RMSE when compared to

**TABLE 1** RMSE of  $\hat{\nu}$  under different methods, with true  $\nu = 2$  and sample size  $n = 300$ .

| $p$ | $\hat{\nu}$ | $\hat{\nu}_{adj}$ | $\hat{\nu}_{Bayes}$ | $\hat{\nu}_{Pseudo}$ |
|-----|-------------|-------------------|---------------------|----------------------|
| 1   | 0.3456      | 0.3512            | 0.3545              | 0.3577               |
| 2   | 0.3441      | 0.3523            | 0.3527              | 0.3558               |
| 5   | 0.3363      | 0.3531            | 0.3438              | 0.3466               |
| 10  | 0.3362      | 0.3655            | 0.3422              | 0.3444               |
| 20  | 0.3381      | 0.3838            | 0.3405              | 0.3411               |
| 40  | 0.4361      | 0.4488            | 0.4322              | 0.4267               |
| 60  | 1.2713      | 0.5810            | 1.2935              | 1.2486               |
| 80  | 1.6681      | 0.9010            | 1.6608              | 1.6751               |

each other and to the likelihood estimators. Other choices of  $\nu$ ,  $n$ , and  $p$ , reported in the supplementary material, give similar conclusions. If  $p$  is not large compared to  $n$ , we recommend the  $\hat{\nu}$  and for larger  $p$ ,  $\hat{\nu}_{adj}$ .

### 3 | ESTIMATING THE REGRESSION

#### 3.1 | Simulation Settings

In linear regression, estimating the coefficient is the primary objective. To evaluate the effect of  $\nu$ -estimation in supporting coefficient estimation, we designed a simulation study using the stack loss data of Brownlee (1965). It has 21 responses ( $Y$ ) in an industrial process, along with three explanatory variables: air flow ( $X_1$ ), temperature ( $X_2$ ), and acid concentration ( $X_3$ ). We conducted simulations according to the following three data generation methods:

1. In the first method, we used the original design matrix  $X$  from the stackloss dataset, which contains  $n = 21$  observations and  $p = 4$  predictors (including an intercept). The least-squares estimates of  $\beta$  and  $\sigma$  for this dataset were treated as the true values. We generated response values according to

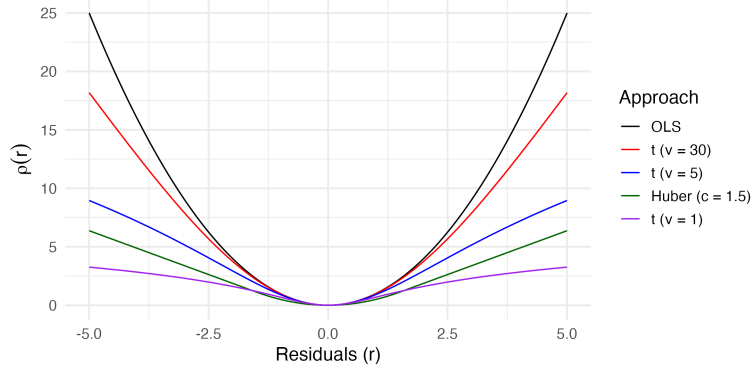
$$y_i = x_i^T \beta + \epsilon_i, \quad i = 1, \dots, 21, \quad (7)$$

with  $X$  from the original data, and  $t_2$  errors  $\epsilon_i$ .

2. In the second method, we expanded the data to  $p = \{40, 80, 120\}$  and  $n = 210$  to study the performance in a higher-dimensional regime with a larger sample size. The first four predictors (including the intercept) were based on the original stackloss predictors, in which we retained the original 21 observations, and generated an additional 189 observations with 3 predictors from a multivariate normal distribution with the same empirical mean vector and covariance matrix obtained from the original design matrix. All the remaining  $p - 4$  predictors were generated independently from a standard normal distribution. The corresponding true  $\beta$  vector had the first four coefficients being the OLS estimates from the original data, and the remaining coefficients set as zero.
3. In the third method, we generated the data using (7), with  $\epsilon_i$  following either  $t_2$  or  $N(0, 1)$  to understand the effect of error distribution. Data was simulated with these  $(n, p)$  settings: (20, 4), (500, 4), and (100, 50).

We considered three approaches for the estimation of  $\nu$ : profile likelihood  $\hat{\nu}_p$ , adjusted profile likelihood  $\hat{\nu}_{adj}$ , and Jeffreys  $\hat{\nu}_{Bayes}$ . We also considered some fixed value other than the true  $\nu = 2$ . For Bayesian methods, we only considered the Jeffreys approach because both Bayesian methods give very similar  $\nu$ -estimation in Section 2, and full independence Jeffreys is easier to implement, as it does not first construct the profile likelihood function. In each simulation, we estimated or fixed  $\nu$  from the four approaches, then obtained  $(\hat{\beta}_\nu, \hat{\sigma}_\nu)$  by optimizing the Student's  $t$  likelihood function. We also considered the ordinary least squares (OLS) and Huber regression estimators (Huber, 1973) of  $\beta$  for comparison. For Huber's method, the tuning parameter was automatically selected using the `rlmDataDriven` R package (Wang et al., 2019). This procedure was repeated over 500 simulations, again keeping the design matrix  $X$  fixed while varying only the error term.

OLS imposes the largest penalty on the outliers compared to other methods.  $t_1$  imposes lighter penalty than Huber regression with tuning parameter  $c = 1.5$ ). As  $\nu$  increases, the loss function behaves more similarly as OLS loss function. See Figure 2.



**FIGURE 2** Loss function of OLS vs Huber vs  $t$  distribution

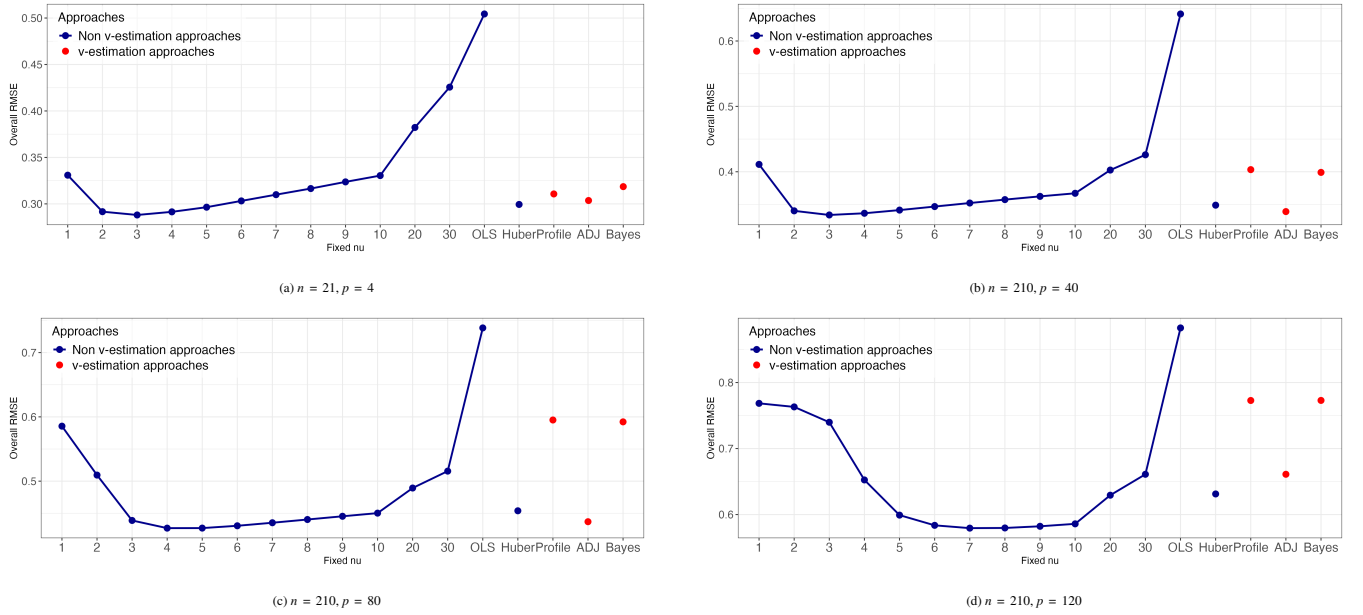
The simulation RMSE for  $\hat{\beta}$  was computed as the average of the square root MSE over the scalar, ignoring the intercept

$$RMSE(\hat{\beta}) = \frac{1}{500} \sum_{k=1}^{500} \sqrt{\frac{1}{p-1} \sum_{j=2}^p (\beta_{jk} - \hat{\beta}_{jk})^2}. \quad (8)$$

where  $\hat{\beta}_{jk}$  is the  $k$ th simulation's estimate of  $j$ th coefficient.

### 3.2 | Results

**FIGURE 3** Overall RMSE of  $\hat{\beta}$  using stackloss data (data generation method 1)

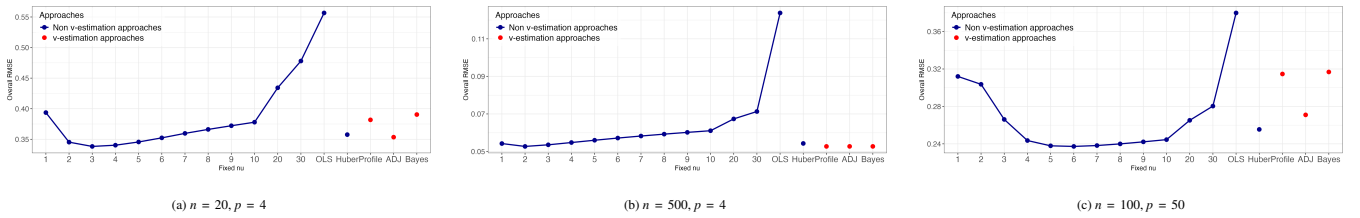


**FIGURE 3** The plots compare fixed  $\nu$  approaches (in blue) for  $\nu \in \{1, \dots, 10\} \cup \{20, 30\}$  and other traditional estimators (OLS and Huber) with the proposed  $\nu$ -estimation approaches (in red).

From Figure 3a, we observe that the lowest overall RMSE is achieved when fixing  $\nu = 3$ , with the RMSE at the true value  $\nu = 2$  is only slightly higher, indicating that the method is not highly sensitive to small misspecifications of  $\nu$  around the true value. The slightly better performance at  $\nu = 3$  compared to the true  $\nu = 2$  is likely due to sampling variability in the simulation. As  $n$  increases, the true  $\nu$  would result in the lowest RMSE. For higher values of  $\nu$  (e.g.,  $\nu = 20$  or  $\nu = 30$ ), the RMSE for the estimate of  $\beta$  increases notably. The three  $\nu$ -estimation approaches are comparable to the best fixed- $\nu$  results, and all outperform ordinary least squares. Estimation of  $\beta$  using  $\hat{\nu}_{adj}$  gives the smallest RMSE among  $\nu$  estimated methods, which is comparable to RMSE of fixed  $\nu = 6$ . Huber regression is slightly better than  $\hat{\nu}_{adj}$  approach.

According to Figures 3b and 3c, when  $p = 40, 80$  ( $p/n \approx 0.190, 0.381$  respectively), the lowest overall RMSE is achieved by fixing  $\nu$  at 3 or 4, and the RMSE increases for  $\nu > 4$ . The  $\hat{\nu}_{adj}$  approach outperforms Huber regression and gives the lowest RMSE among all  $\nu$  estimated methods, which is comparable to fixing at the true  $\nu$ . In Figure 3d, as we further increase  $p$  to 120 ( $p/n \approx 0.571$ ), the lowest overall RMSE is achieved at larger fixed  $\nu = 7$ . High dimensionality of  $\beta$  gives the model enough flexibility to absorb extreme values into the fitted coefficients, shrinking the residuals relative to the true  $t_2$  errors. With these outliers muted, the likelihood favours a larger degrees of freedom  $\nu$ , similar to selecting a less robust (steeper) loss function in Figure 2, since the data no longer exhibit the heavy-tailed behaviour that would justify a small  $\nu$ . Among the  $\nu$ -estimation approaches,  $\hat{\nu}_{adj}$  approach performs the best, its RMSE value is relatively large compared to the lowest RMSE that could be achieved by traditional methods. In particular, Huber regression performs better than the  $\hat{\nu}_{adj}$  approach.

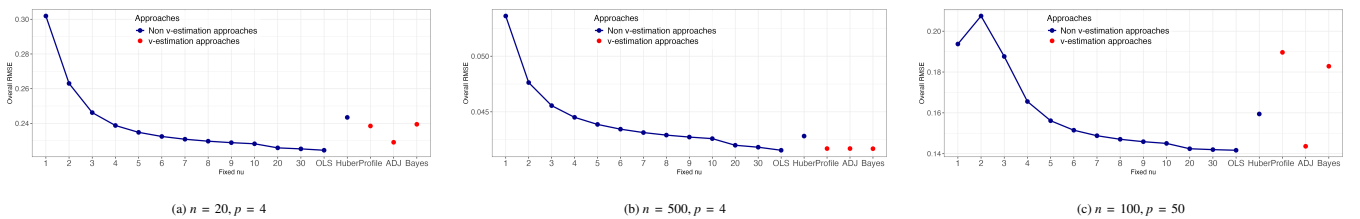
**FIGURE 4** Overall RMSE of  $\hat{\beta}$  in simulated  $t$ -error data (data generation method 2)



**FIGURE 4** The plots compare non  $\nu$ -estimation approaches (in blue) across various fixed values of  $\nu$  and other traditional estimators (OLS and Huber) with the proposed  $\nu$ -estimation approaches (in red) in terms of overall RMSE.

According to Figures 4a and 4b, the trend in overall RMSE using the simulated data with  $t$  error is similar to that of the stack loss data when  $p/n < 0.5$ , where RMSE tends to increase as the fixed  $\nu$  shifts away from the true value. The  $\hat{\nu}_{adj}$  approach results in a lower RMSE than Huber regression. According to Figure 4c, when  $p/n = 0.5$ , we observe a similar RMSE trend as the stackloss  $n = 210, p = 120$  simulations. The lowest RMSE is achieved at fixed  $\nu = 6$ , which is not close to true  $\nu = 2$ . Huber regression performs better than the  $\hat{\nu}_{adj}$  approach, although both approaches give relatively large RMSEs compared to the smallest RMSE given by fixing  $\nu = 6$ .

**FIGURE 5** Overall RMSE of  $\hat{\beta}$  using simulated normal-error data (data generation method 3)



**FIGURE 5** The plot compares non  $\nu$ -estimation approaches (in blue) across various fixed values of  $\nu$  and other traditional estimators (OLS and Huber) with the proposed  $\nu$ -estimation approaches (in red) in terms of overall RMSE.

With standard normal errors (Figures 5a and 5b), OLS performs the best in both  $n = 20$  and  $n = 500$  cases. All  $\nu$ -estimation approaches outperform Huber regression with  $\hat{\nu}_{adj}$  approach coming closest to OLS among the  $\nu$ -estimation approaches. In

the high-dimensional case  $n = 100; p = 50$ , Figure 5c shows that OLS again performs the best and only  $\hat{\nu}_{adj}$  surpasses Huber regression among the  $\nu$ -estimation approaches. Overall, estimating  $\nu$  is beneficial when  $p/n$  is not too large, as it would give promising results regardless the underlying error distribution. For errors with light-tail distribution, adjusted profile likelihood approach is comparable to the OLS, and for heavy tail distribution, it outperforms Huber when  $p/n$  ratio is below 0.5. Although Lange et al. (1989) recommends fixing  $\nu$  for small sample size  $n$ , the adjusted profile likelihood approach yields performance comparable to the best fixed- $\nu$  choices across error settings, supporting its use for estimating  $\nu$ .

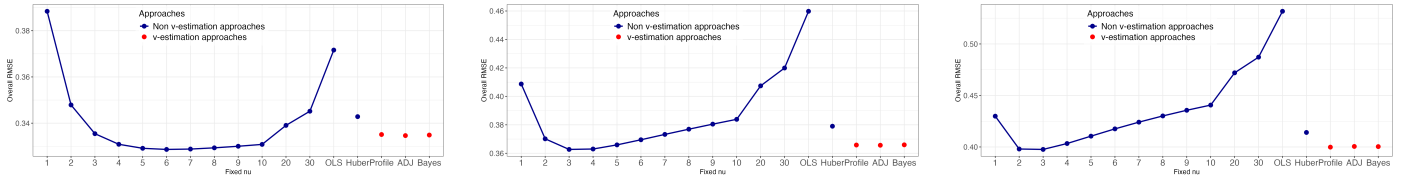
## 4 | ROBUSTNESS TO OUTLIERS

### 4.1 | Simulation Settings

To investigate  $t$  regression on data with outliers, we considered a linear model with true  $\beta = (1, 5, 10)$  and  $n = 100$ . Additionally, we also considered data with outliers under high dimensional setting, generated using a linear model with  $p = 80$ , true  $\beta = \mathbf{0}_p$ , and  $n = 300$ . The covariates and errors were generated from the standard normal distribution, with contamination (i)  $N(0, 9)$ , (ii)  $\chi^2(4) - 4$ , (iii)  $t(2)$ , or (iv) two-point contamination in which any response value may become  $-5$  or  $5$  with a probability  $\lambda/2$  each. The contamination rates considered are  $\lambda = 10\%$ ,  $20\%$ , and  $30\%$ .

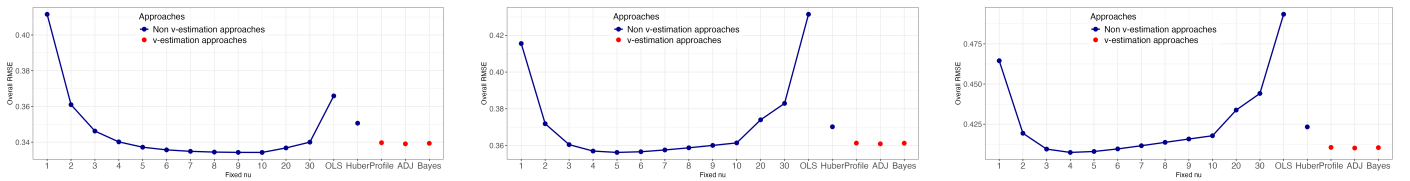
### 4.2 | Results when $p = 3$

**FIGURE 6** Overall RMSE of  $\hat{\beta}$  with 10%, 20%, and 30%  $N(0, 9)$  contaminated errors when  $p = 3$



When the data is contaminated with  $N(0, 9)$  errors (see Figure 6), although the  $\nu$ -estimation approaches do not necessarily give the lowest overall RMSE, they are able to give relatively low RMSE compared to OLS and Huber regression. As the contamination rate increases, the  $\nu$ -estimation approaches outperform most of the fixed  $\nu$  approach estimations as they are successful in estimating  $\nu$  close to truth. Similar pattern is shown in data with two-point contamination.

**FIGURE 7** Overall RMSE of  $\hat{\beta}$  with 10% 20%, and 30%  $\chi^2(4) - 4$  contaminated errors when  $p = 3$

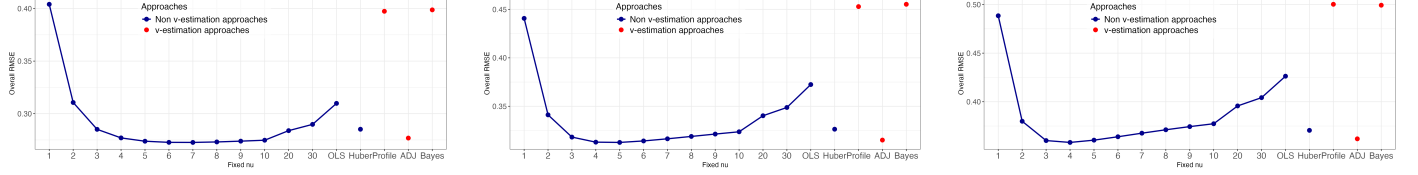


In cases where data is contaminated with  $\chi^2(4) - 4$  (see Figure 7) or  $t(2)$  errors, the  $\nu$ -estimation approaches outperform OLS and Huber regression. Regardless of the contamination rates, a moderate amount of fixed  $\nu$ s give slightly lower overall RMSE than all  $\nu$ -estimation approaches.

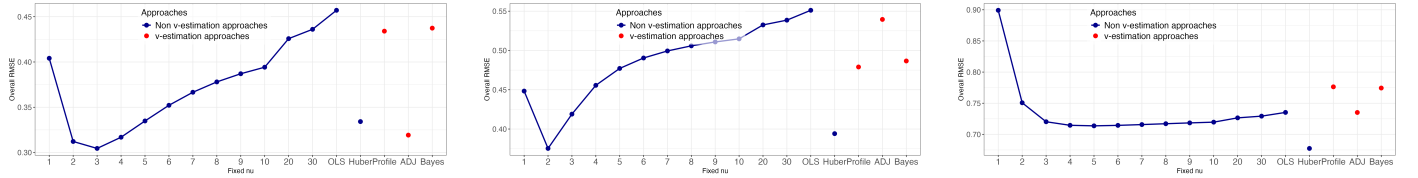
### 4.3 | Results when $p = 80$

In high dimension setting, when the data is contaminated with  $N(0, 9)$  (see Figure 8),  $\chi^2(4) - 4$ , or  $t(2)$  errors, although  $\hat{\nu}_{adj}$  approach does not necessarily give the lowest overall RMSE, it is able to give relatively lower RMSE compared to OLS and Huber regression. This may be explained by the fact that  $\hat{\nu}_{adj}$  are closer to the truth compared to other  $\nu$  estimation methods (and fixing large  $\nu$ ) in high dimension setting.

**FIGURE 8** Overall RMSE of  $\hat{\beta}$  with 10%, 20%, and 30%  $N(0, 9)$  contaminated errors when  $p = 80$



**FIGURE 9** Overall RMSE of  $\hat{\beta}$  with 10%, 20%, and 30% two-point contaminated errors when  $p = 80$



For data contaminated with two-point errors (see Figure 9),  $\hat{\nu}_{adj}$  approach outperforms Huber and OLS when the contamination rate is low (e.g. 10%). As the contamination rate increases, Huber regression performs better as  $\hat{\nu}_{adj}$  are overestimated, and hence the corresponding  $\beta$  estimation performance is similar to fixing  $\nu = 30$ . Nonetheless, the  $\nu$ -estimation approaches outperform OLS when the contamination rate is 20%. As the contamination rate increases to 30%, all  $\nu$ -estimation approaches perform as bad as most fixed- $\nu$  approaches and OLS. This may be caused by high contamination of outliers with a "uniform" pattern in the data that t distribution fails to capture.

These results collectively suggest that the  $\nu$ -estimation approaches, and  $\hat{\nu}_{adj}$  in some high dimension settings, offer a robust performance across various contamination types and levels, often outperforming classical regression methods.

## 5 | CONCLUSIONS AND DISCUSSION

In this paper, we studied robust regression with Student's t error and addressed the commonly asked question on how to choose the degree of freedom parameter in Student's t distribution in practice. We compared 4 frequentist and Bayesian methods in estimating the degree of freedom  $\nu$ , and studied how these choices propagate to estimation of the regression coefficients  $\beta$  in uncontaminated and contaminated data. In terms of  $\beta$  estimation performance, we suggest estimating  $\nu$  with adjusted profile likelihood when the  $p/n$  ratio is not too large as it is the most robust in both heavy-tailed and light-tailed distribution, as well as in contaminated data and large  $p$  (given moderate  $n$ ) settings. However, to ensure stable  $\nu$  and  $\beta$  estimations, our proposed  $\nu$ -estimation methods require  $p$  to be not too large relative to  $n$ . This is because high dimensionality may lead to larger residuals magnitude and tend to overestimate  $\nu$ . Therefore, when  $p/n$  ratio is very large (it would be better if we were able to quantify this), optimizing the likelihood by fixing  $\nu$  at some large values may be more efficient.



## AUTHOR CONTRIBUTIONS

This is an author contribution text.

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## FINANCIAL DISCLOSURE

None reported.

## CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

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